



OMNIKON

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TEAM NAME

SPARKLE

MEMBERS

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THEME

Reliable Distress Alerts for Women's Safety

PROBLEM STATEMENT

Existing safety apps depend on continuous internet and multiple deliberate taps. In a real emergency — panic, poor network, or a locked phone — alerts are delayed or never sent, especially in rural and low-connectivity areas.

Proposed Solution (Idea Overview)

- **Multi-trigger SOS — 5x power-button press, a BLE wearable button, or the voice keyword “Help” — works even with the screen locked**
- **3-tier fallback delivery: live alert over internet → auto-SMS over 2G if data fails → Bluetooth mesh relay via nearby SPARKLE users if there’s no signal at all**
- **Silent, disguised mode — shows a fake incoming-call screen so an aggressor never suspects an alert was sent**
- **Breadcrumb safe-journey trail — shares live location with trusted contacts on timed journeys, auto-escalating if a check-in is missed**

○ *Technologies to be used*

Android (Kotlin) app • ESP32 / nRF52 BLE wearable band • Node.js + Firebase backend • Twilio / telecom SMS gateway • BLE mesh relay layer • Maps API • 112 India / Dial-100 API bridge

○ *Methodology and process for implementation*

- 1. Trigger detected (button / BLE / voice)**
- 2. Internet available → live app alert + audio sent to contacts & police dashboard**
- 3. No internet → automatic SMS with GPS coordinates (2G fallback)**
- 4. No signal → Bluetooth mesh relay through nearby SPARKLE phones**
- 5. Alert confirmed → contacts & control room notified**

○ *Analysis of the feasibility of the idea*

- **Runs on any existing smartphone — no new device required for the core app**
- **SMS fallback works on basic 2G, reaching rural / low-connectivity users**
- **Wearable BOM under ₹ 500; reuses existing 112 India / Dial-100 infrastructure**

○ *Potential challenges and risks*

- **False alarms, battery drain on the wearable, sparse mesh coverage in low-density areas**
- **Data privacy concerns and cost of wearable adoption at scale**

○ *Strategies for overcoming these challenges*

- **10-second cancel window + adaptive duty-cycling for battery (5+ days standby)**
- **Phased pilot in dense zones first (campuses, transit); SMS remains guaranteed fallback**
- **End-to-end encrypted, opt-in location data with auto-delete; subsidized wearables via NGO/police partners**

○ *Potential impact on the target audience*

Gives women in both urban high-density and rural low-connectivity areas a genuinely reliable way to call for help — restoring confidence to commute, work late, and travel independently, without depending on always-on data.

○ *Benefits of the solution*

- **Social:** Faster response reduces harm and builds trust in public spaces.
- **Economic:** Low BOM cost + reuse of existing 112 infrastructure keeps city-wide rollout cheap.
- **Environmental / Governance:** Structured alert data helps police & NGOs identify unsafe hotspots and target resources.

- **Government of India — 112 India / Dial-100 Emergency Response Support System (ERSS), used as the escalation endpoint for verified alerts**
- **Comparative review of existing personal-safety apps (bSafe, Raksha, Himmat Plus) — core gap identified: all rely on continuous internet, motivating the 3-tier fallback design**
- **Bluetooth mesh relay networking for offline communication — basis for the peer-to-peer alert propagation layer in signal-dead zones**
- **Ministry of Women & Child Development — Nirbhaya Fund safety-technology guidelines, referenced for scalable public-safety integration**
- **Studies on SMS-based emergency alerting over 2G networks in low-connectivity regions — informing the guaranteed-delivery fallback tier**